## ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **Perimeter and Area** | Class 6 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** 162 | Let breadth $= b$. |
| 2 | **(a)** 16 | Perimeters are equal. |
| 3 | **(d)** 120 | Half of the rectangle. |
| 4 | **(b)** 336 | Outer minus inner. |
| 5 | **(c)** 14 | Perimeter does not change. |
| 6 | **(d)** 5 | Area = cost ÷ rate; breadth = area ÷ length. |
| 7 | **(b)** 64 | A square has the greatest area for a given perimeter. |
| 8 | **(b)** 42 | Add the three sides. |
| 9 | **(a)** 84 | Outer side $= a + 2$. |
| 10 | **(c)** 147 | Let breadth $= b$. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | $l + b = 30$; ratio parts $= 5$ (1); Length $= 18$ cm, breadth $= 12$ cm (1). | 1+1 |
| 12 | Area $= 1200$ m$^2$, perimeter $= 140$ m; Levelling cost $= 6000$ (1); Fencing cost $= 1680$; total $= 7680$ (1). | 1+1 |
| 13 | Room area $= 48$ m$^2$; Carpet area $= 20$ m$^2$ (1); Uncovered $= 28$ m$^2$ (1). | 1+1 |
| 14 | $l + b = 32$; ratio parts $= 8$ (1); Length $= 20$ cm, breadth $= 12$ cm (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Perimeter $= 130$ m; Wire needed $= 3 \times 130 = 390$ m (1); Fencing cost $= 4680$ (1); Ploughing cost $= 4000$ (1). | 1+1+1 |
| 16 | Square perimeter $= 4 \times 12 = 48$ cm; rectangle perimeter $= 2(14 + 10) = 48$ cm; Square area $= 144$ cm$^2$ (1); Rectangle area $= 140$ cm$^2$ (1); Square is larger by $4$ cm$^2$ (1). | 1+1+1 |
| 17 | Floor in cm: $600$ by $400$; Tile area $600$ cm$^2$ (1); Floor area $240000$ cm$^2$ (1); Tiles $= 400$ (1). | 1+1+1 |
| | **OR:** Perimeter $= 160$ m; Wire needed $= 2 \times 160 = 320$ m (1); Fencing cost $= 4800$ (1); Ploughing cost $= 7500$ (1). | 1+1+1 |
| 18 | Square perimeter $= 4 \times 15 = 60$ cm; rectangle perimeter $= 2(18 + 12) = 60$ cm; Square area $= 225$ cm$^2$ (1); Rectangle area $= 216$ cm$^2$ (1); Square is larger by $9$ cm$^2$ (1). | 1+1+1 |
| | **OR:** Floor in cm: $800$ by $500$; Tile area $1000$ cm$^2$ (1); Floor area $400000$ cm$^2$ (1); Tiles $= 400$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Rectangle area $= 240$ cm$^2$ (1); Triangle area $= \frac{1}{2} \times 20 \times 8 = 80$ cm$^2$ (1); Remaining $= 160$ cm$^2$ (1); Fraction left $= \frac{160}{240}$ (2). | 1+1+1+2 |
| | **OR:** Path parallel to length: $60 \times 3 = 180$ m$^2$ (1); Path parallel to breadth: $40 \times 3 = 120$ m$^2$ (1); Common square $= 3 \times 3 = 9$ m$^2$ (1); Area of paths $= 291$ m$^2$ (1); Rest of the park $= 2109$ m$^2$ (1). | 1+1+1+1+1 |
| 20 | Rectangle area $= 240$ cm$^2$ (1); Triangle area $= \frac{1}{2} \times 24 \times 6 = 72$ cm$^2$ (1); Remaining $= 168$ cm$^2$ (1); Fraction left $= \frac{168}{240}$ (2). | 1+1+1+2 |
| | **OR:** Path parallel to length: $80 \times 4 = 320$ m$^2$ (1); Path parallel to breadth: $50 \times 4 = 200$ m$^2$ (1); Common square $= 4 \times 4 = 16$ m$^2$ (1); Area of paths $= 504$ m$^2$ (1); Rest of the park $= 3496$ m$^2$ (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Perimeter and Area** | Class 6 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** 20 | Perimeters are equal. |
| 2 | **(c)** 240 | Half of the rectangle. |
| 3 | **(a)** 584 | Outer minus inner. |
| 4 | **(b)** 18 | Perimeter does not change. |
| 5 | **(b)** 7 | Area = cost ÷ rate; breadth = area ÷ length. |
| 6 | **(c)** 100 | A square has the greatest area for a given perimeter. |
| 7 | **(a)** 68 | Add the three sides. |
| 8 | **(c)** 124 | Outer side = $a + 2$. |
| 9 | **(b)** 100 | Let breadth = $b$. |
| 10 | **(c)** 28 | Perimeters are equal. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Area = 1000 m$^2$, perimeter = 140 m; Levelling cost = 4000 (1); Fencing cost = 2100; total = 6100 (1). | 1+1 |
| 12 | Room area = 63 m$^2$; Carpet area = 30 m$^2$ (1); Uncovered = 33 m$^2$ (1). | 1+1 |
| 13 | $l + b = 49$; ratio parts = 7 (1); Length = 28 cm, breadth = 21 cm (1). | 1+1 |
| 14 | Area = 2700 m$^2$, perimeter = 210 m; Levelling cost = 8100 (1); Fencing cost = 2100; total = 10200 (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Perimeter = 190 m; Wire needed = $3 \times 190 = 570$ m (1); Fencing cost = 5700 (1); Ploughing cost = 6300 (1). | 1+1+1 |
| 16 | Square perimeter = $4 \times 20 = 80$ cm; rectangle perimeter = $2(24 + 16) = 80$ cm; Square area = 400 cm$^2$ (1); Rectangle area = 384 cm$^2$ (1); Square is larger by 16 cm$^2$ (1). | 1+1+1 |
| 17 | Floor in cm: 900 by 600; Tile area 450 cm$^2$ (1); Floor area 540000 cm$^2$ (1); Tiles = 1200 (1). | 1+1+1 |
| | **OR:** Perimeter = 130 m; Wire needed = $4 \times 130 = 520$ m (1); Fencing cost = 4160 (1); Ploughing cost = 5400 (1). | 1+1+1 |
| 18 | Square perimeter = $4 \times 18 = 72$ cm; rectangle perimeter = $2(23 + 13) = 72$ cm; Square area = 324 cm$^2$ (1); Rectangle area = 299 cm$^2$ (1); Square is larger by 25 cm$^2$ (1). | 1+1+1 |
| | **OR:** Floor in cm: 1000 by 400; Tile area 1000 cm$^2$ (1); Floor area 400000 cm$^2$ (1); Tiles = 400 (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Rectangle area = 420 cm$^2$ (1); Triangle area = $\frac{1}{2} \times 30 \times 10 = 150$ cm$^2$ (1); Remaining = 270 cm$^2$ (1); Fraction left = $\frac{270}{420}$ (2). | 1+1+1+2 |
| | **OR:** Path parallel to length: $70 \times 2 = 140$ m$^2$ (1); Path parallel to breadth: $45 \times 2 = 90$ m$^2$ (1); Common square = $2 \times 2 = 4$ m$^2$ (1); Area of paths = 226 m$^2$ (1); Rest of the park = 2924 m$^2$ (1). | 1+1+1+1+1 |
| 20 | Rectangle area = 144 cm$^2$ (1); Triangle area = $\frac{1}{2} \times 16 \times 6 = 48$ cm$^2$ (1); Remaining = 96 cm$^2$ (1); Fraction left = $\frac{96}{144}$ (2). | 1+1+1+2 |
| | **OR:** Path parallel to length: $90 \times 5 = 450$ m$^2$ (1); Path parallel to breadth: $60 \times 5 = 300$ m$^2$ (1); Common square = $5 \times 5 = 25$ m$^2$ (1); Area of paths = 725 m$^2$ (1); Rest of the park = 4675 m$^2$ (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679


## ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Perimeter and Area** | Class 6 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** 252 | Half of the rectangle. |
| 2 | **(d)** 252 | Outer minus inner. |
| 3 | **(c)** 21 | Perimeter does not change. |
| 4 | **(a)** 6 | Area = cost ÷ rate; breadth = area ÷ length. |
| 5 | **(a)** 144 | A square has the greatest area for a given perimeter. |
| 6 | **(a)** 84 | Add the three sides. |
| 7 | **(c)** 363 | Let breadth = $b$. |
| 8 | **(d)** 36 | Perimeters are equal. |
| 9 | **(a)** 360 | Half of the rectangle. |
| 10 | **(d)** 900 | Outer minus inner. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Room area = 60 m$^2$; Carpet area = 28 m$^2$ (1); Uncovered = 32 m$^2$ (1). | 1+1 |
| 12 | $l + b = 36$; ratio parts = 12 (1); Length = 21 cm, breadth = 15 cm (1). | 1+1 |
| 13 | Area = 875 m$^2$, perimeter = 120 m; Levelling cost = 5250 (1); Fencing cost = 960; total = 6210 (1). | 1+1 |
| 14 | Room area = 108 m$^2$; Carpet area = 48 m$^2$ (1); Uncovered = 60 m$^2$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Perimeter = 220 m; Wire needed = $2 \times 220 = 440$ m (1); Fencing cost = 8800 (1); Ploughing cost = 5600 (1). | 1+1+1 |
| 16 | Square perimeter = $4 \times 25 = 100$ cm; rectangle perimeter = $2(31 + 19) = 100$ cm; Square area = 625 cm$^2$ (1); Rectangle area = 589 cm$^2$ (1); Square is larger by 36 cm$^2$ (1). | 1+1+1 |
| 17 | Floor in cm: 1200 by 500; Tile area 1200 cm$^2$ (1); Floor area 600000 cm$^2$ (1); Tiles = 500 (1). | 1+1+1 |
| | **OR:** Perimeter = 170 m; Wire needed = $3 \times 170 = 510$ m (1); Fencing cost = 7140 (1); Ploughing cost = 6600 (1). | 1+1+1 |
| 18 | Square perimeter = $4 \times 16 = 64$ cm; rectangle perimeter = $2(23 + 9) = 64$ cm; Square area = 256 cm$^2$ (1); Rectangle area = 207 cm$^2$ (1); Square is larger by 49 cm$^2$ (1). | 1+1+1 |
| | **OR:** Floor in cm: 700 by 600; Tile area 700 cm$^2$ (1); Floor area 420000 cm$^2$ (1); Tiles = 600 (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Rectangle area = 144 cm$^2$ (1); Triangle area = $\frac{1}{2} \times 18 \times 4 = 36$ cm$^2$ (1); Remaining = 108 cm$^2$ (1); Fraction left = $\frac{108}{144}$ (2). | 1+1+1+2 |
| | **OR:** Path parallel to length: $100 \times 4 = 400$ m$^2$ (1); Path parallel to breadth: $70 \times 4 = 280$ m$^2$ (1); Common square = $4 \times 4 = 16$ m$^2$ (1); Area of paths = 664 m$^2$ (1); Rest of the park = 6336 m$^2$ (1). | 1+1+1+1+1 |
| 20 | Rectangle area = 420 cm$^2$ (1); Triangle area = $\frac{1}{2} \times 28 \times 10 = 140$ cm$^2$ (1); Remaining = 280 cm$^2$ (1); Fraction left = $\frac{280}{420}$ (2). | 1+1+1+2 |
| | **OR:** Path parallel to length: $50 \times 2 = 100$ m$^2$ (1); Path parallel to breadth: $30 \times 2 = 60$ m$^2$ (1); Common square = $2 \times 2 = 4$ m$^2$ (1); Area of paths = 156 m$^2$ (1); Rest of the park = 1344 m$^2$ (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.